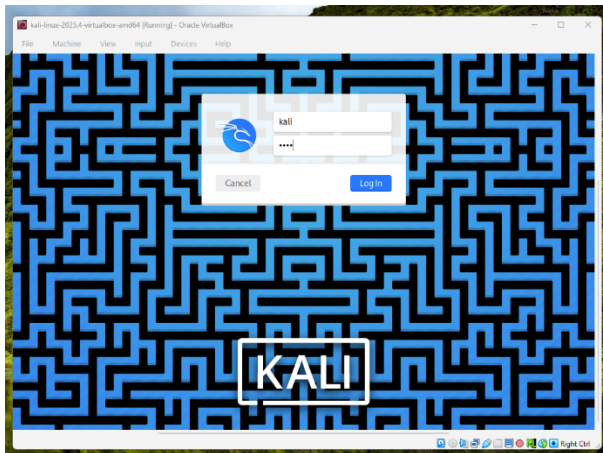


Practical 01

Important: This practical is to be performed strictly in a controlled virtual environment (Kali Linux VM). DVWA (Damn Vulnerable Web Application) is intentionally vulnerable and must not be exposed to any public network or internet-facing system. DVWA is a deliberately insecure web application used for learning web security vulnerabilities such as SQL Injection and Cross Site Scripting.

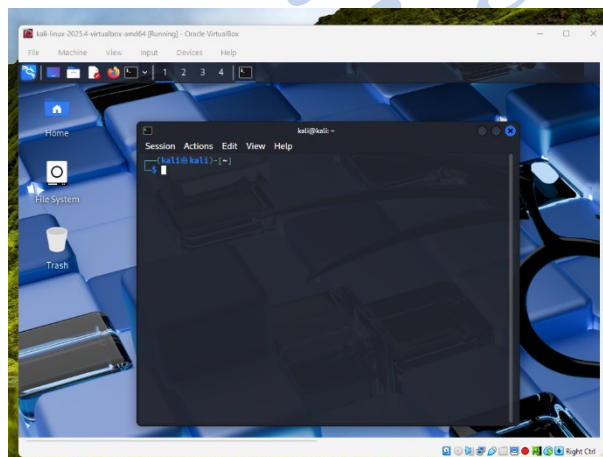
Steps to follow:

1. Log-in Kali



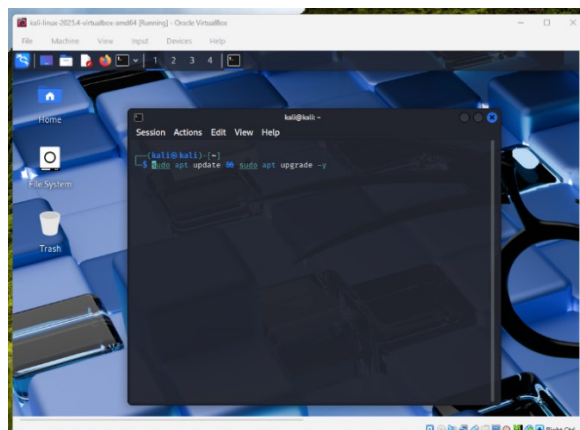
2. Update Kali

a. Open Terminal:



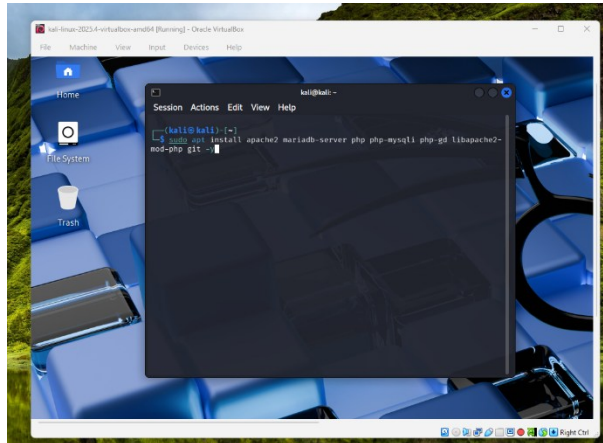
b. Run the following command:

sudo apt update && sudo apt upgrade -y



- c. Run the following command:

```
sudo apt install apache2 mariadb-server php php-mysql php-gd libapache2-mod-php git -y
```

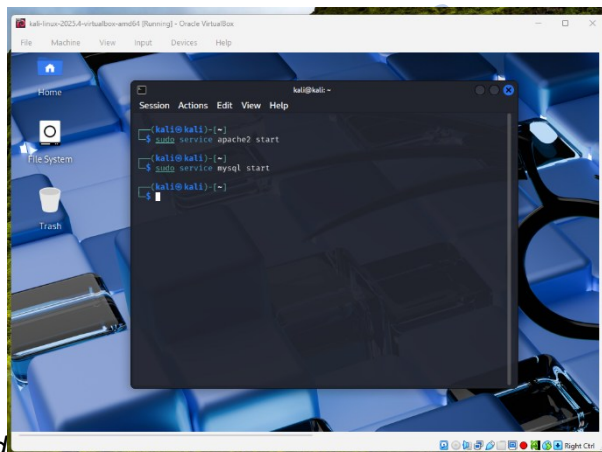


3. Start Services

- a. Run the following commands:

```
sudo service apache2 start
```

```
sudo service mysql start
```



Expected: Apache service should start without errors.

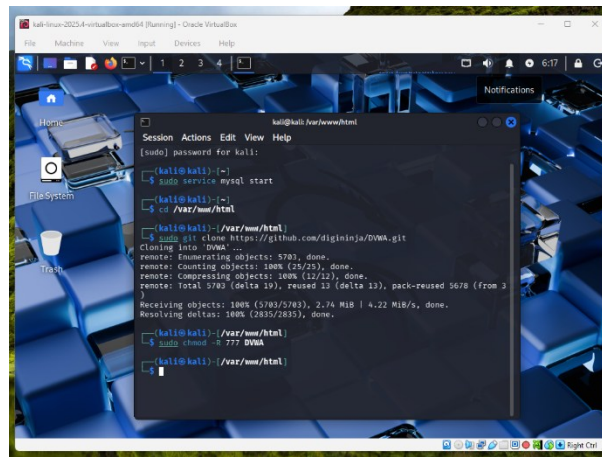
4. Download DVWA

- a. Run the following command:

i. `cd /var/www/html`

ii. `sudo git clone https://github.com/digininja/DVWA.git`

iii. `sudo chmod -R 777 DVWA`

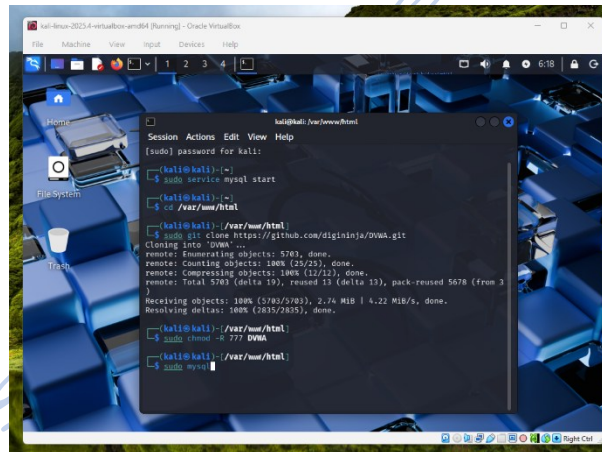


Note: This '777' permission is used only for lab demonstration purposes to avoid permission issues. In real-world environments, secure permissions should be applied.

5. Create Database

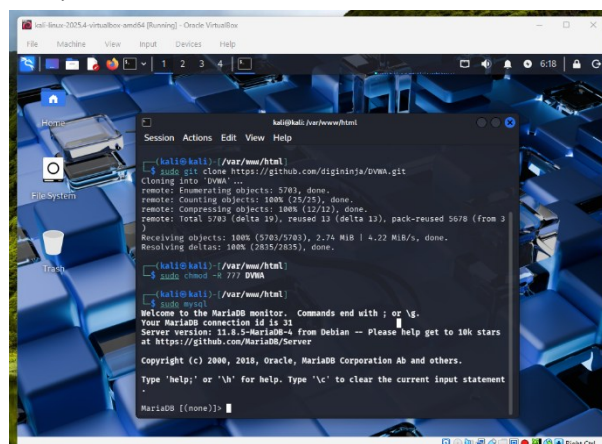
a. Open MySQL:

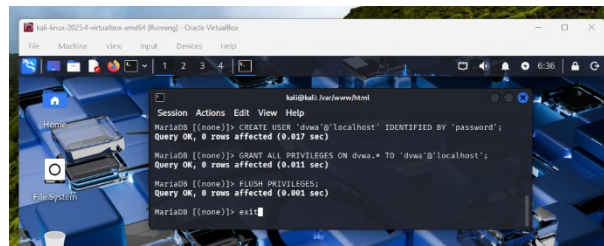
i. `sudo mysql`



b. Run the following commands:

- i. `CREATE DATABASE dvwa;`
- ii. `CREATE USER 'dvwa'@'localhost' IDENTIFIED BY 'password';`
- iii. `GRANT ALL PRIVILEGES ON dvwa.* TO 'dvwa'@'localhost';`
- iv. `FLUSH PRIVILEGES;`
- v. `EXIT;`

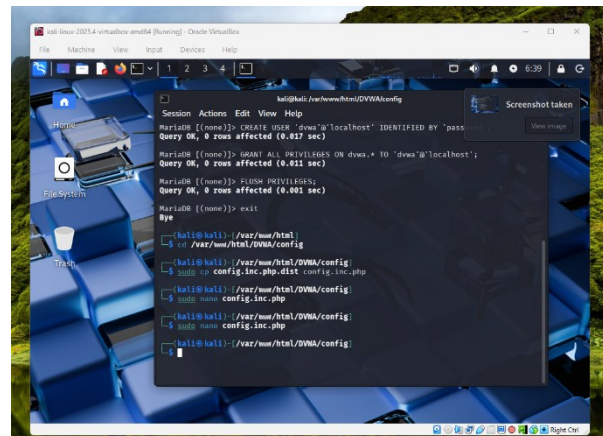




6. Configure DVWA

a. Run the following command:

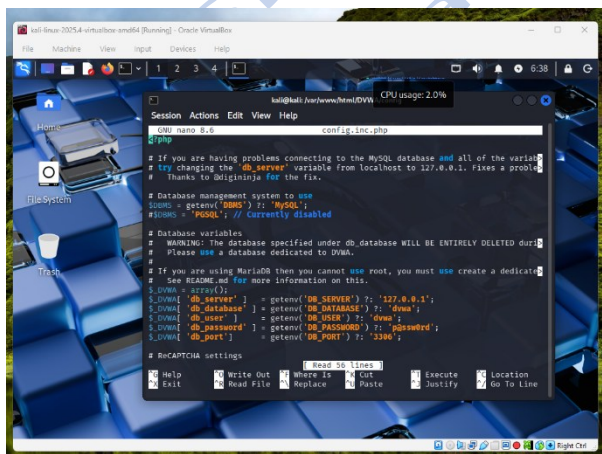
- i. `cd /var/www/html/DVWA/config`
- ii. `sudo cp config.inc.php.dist config.inc.php`
- iii. `sudo nano config.inc.php`



7. Change:

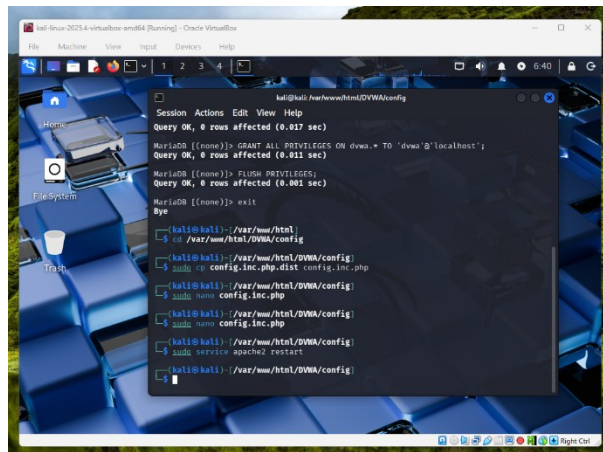
- a. `$_DVWA['db_user'] = 'dwa';`
- b. `$_DVWA['db_password'] = 'password';`

Save and exit.

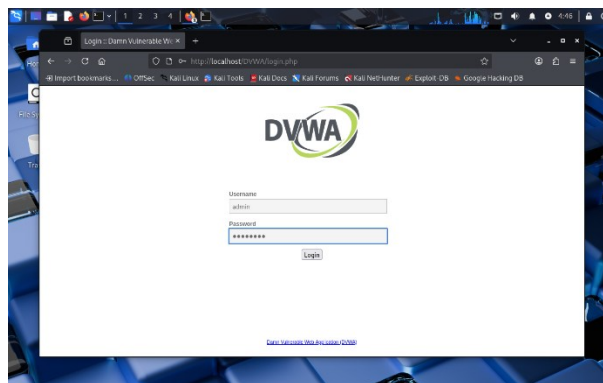


8. Restart Apache

- a. `sudo service apache2 restart`



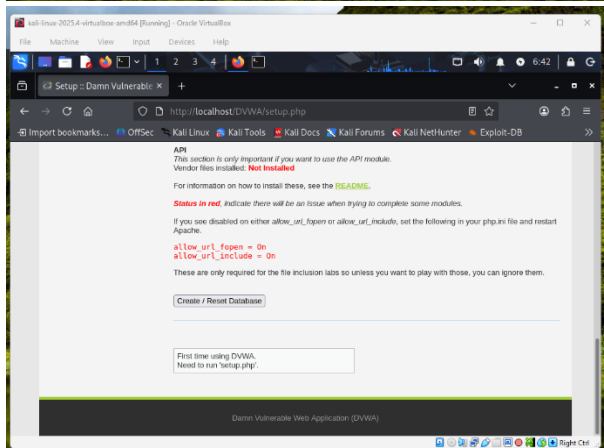
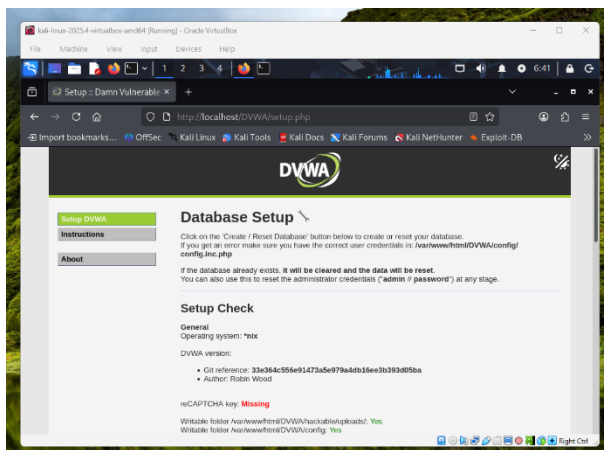
9. Open Browser. Go to:
 - a. *http://localhost/DVWA*



Expected: DVWA login page should appear in browser.

10. Login
 - a. *Username: admin*
 - b. *Password: password*

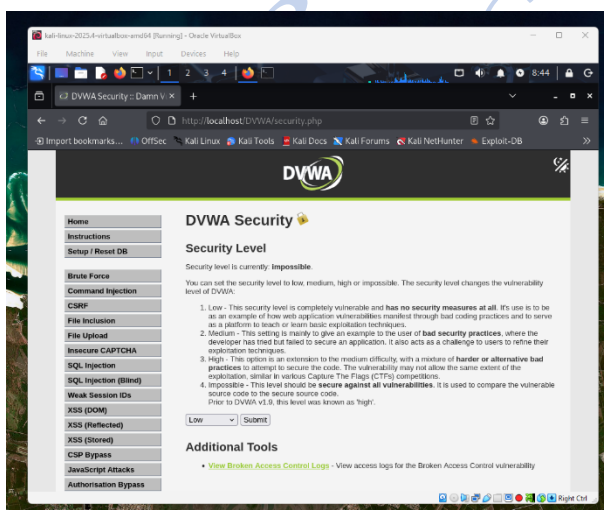
Click Create / Reset Database



11. Set Security Level. Click DVWA Security

a. Select: LOW

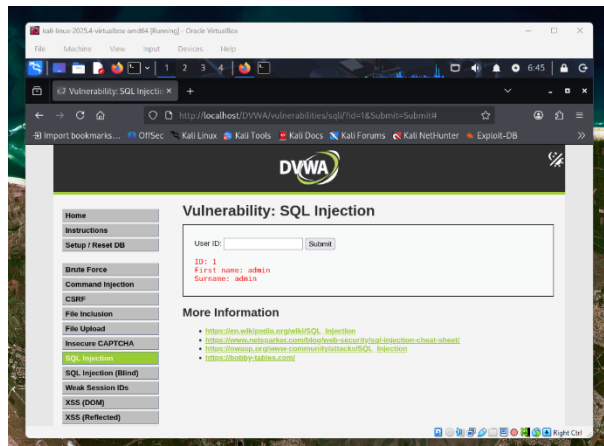
Click Submit.



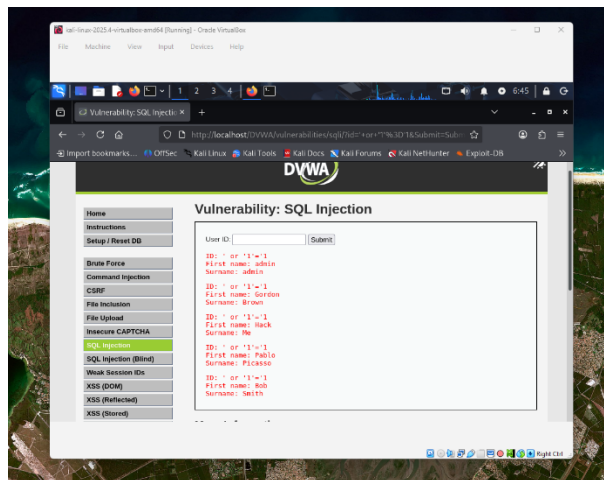
Note: Security level is set to LOW to intentionally allow vulnerabilities for learning purposes.

12. Open SQL Injection page. Click SQL Injection from left menu.

a. In Normal Input, enter: 1. Click Submit to observe the result. Observe single user.



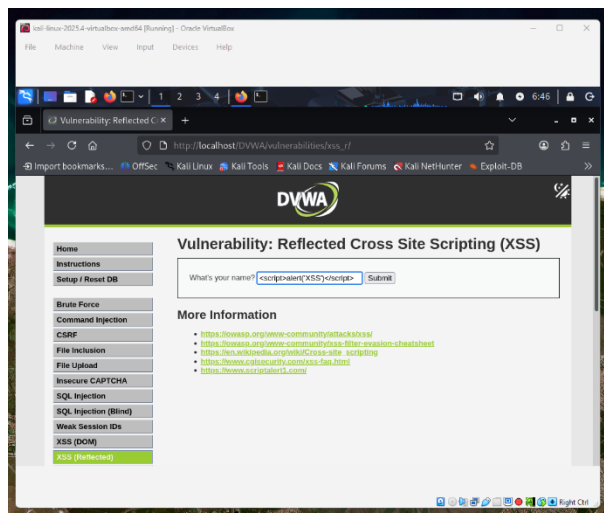
- b. Enter the following input: ' OR '1'='1. Click Submit to observe the result. Observe multiple user records displayed. This input makes the SQL condition always true, resulting in all records being displayed.



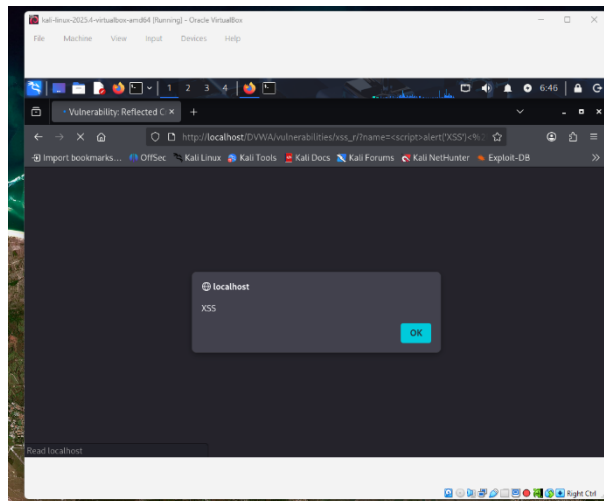
Expected: Multiple user records will be displayed.

13. Open XSS (Reflected): Click XSS (Reflected).

- a. In Normal Input, enter the following input: K. Click Submit.
- b. In XSS Attack, enter the following input: `<script>alert('XSS')</script>`. Click Submit to observe the result. This script is executed in the browser, demonstrating a Cross Site Scripting vulnerability.



14. Popup appears.



15. In the end, run the following command:

- `sudo service apache2 stop`
- `sudo service mysql stop`
- `sudo ip -s -s neigh flush all`

